

Normalization in DBMS is a database design process that organizes data into tables to minimize redundancy, prevent data anomalies (insertion, update, deletion), and ensure data consistency.

Embedded Document

Document ke andar ek aur document.



```
1  {  
2      name: "Manas",  
3      age: 21,  
4      address: {  
5          city: "Bhagalpur",  
6          state: "Bihar",  
7          pincode: 812001  
8      }  
9  }
```

Real World Example:

Agar user ka address sirf user se related hai, to alag collection banane ki zarurat nahi. Isliye MongoDB me hum usko embed kar dete hain.

```
{  
  name: "Rahul",  
  email: "rahul@gmail.com",  
  address: {  
    city: "Delhi",  
    state: "Delhi",  
    country: "India"  
  }  
}
```

Embedded Field Query

Find user by city



```
1 db.users.find({ "address.city": "Delhi" })
```

MongoDB me dot notation use hota hai

Embedded Field Projection

```
{
  name: "Aman",
  address: {
    city: "Delhi",
    pincode: 110001
  }
}

// QUERY:
db.users.find({}, { "address.city": 1 })

// OUTPUT:
{
  _id: ...,
  address: { city: "Delhi" }
}
```

Embedded Field Update

city update

```
db.users.updateOne(  
  { name: "Rahul" },  
  { $set: { "address.city": "Mumbai" } }  
)
```

Embedded Documents Kab Use Karna Hai?

Data tightly related ho (user → address)

Data mostly sath me read hota ho (order → items)

Data chhota ho

Kab Use Nahi Karna Hai?

Jab data bahut bada ho

Jab data multiple documents se related ho

Example: Student & Courses

Ek student multiple courses me ho skta hai, is case me referencing better hai

Referencing

Data ko alag collections me store karna and ID se link kar dena.

```
// users
{
  _id: 1,
  name: "Rahul",
  orders: [1]
}

// orders
{
  productId: 1,
  name: "Laptop",
  price: 50000
}
```


Difference:

Feature	Embedded	Referencing
Speed	Fast (single query)	Slower (multiple query)
Data duplication	Yes	No
Scalability	Limited	High
Update	Difficult	Easy
Use case	Small related data	Large / growing data

Simple Rules

Rule 1: Data size small ho → Embed

Example: User profile, Address

Rule 2: Data frequently update hota ho → Reference

Example: Product price change hota rehta hai

Rule 3: Data huge ho sakta hai → Reference

Example: Comments in YouTube video

Rule 4: Read-heavy system → Embed

Example: Instagram post + few comments

Rule 5: Write-heavy system → Reference

**Thank
You**